JEE MAIN / PREVIOUS YEAR PAPER

JEE Main 2026 21st January Morning Shift

Question Paper with Answer Key

75	ENGLISH	DARK
QUESTIONS	LANGUAGE	FORMAT

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Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

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Consider the following reactions.

$$\underset{\text{(Hot solution)}}{\mathrm{PbCl_2}} + \mathrm{K_2CrO_4} \rightarrow A + 2\mathrm{KCl}$$

$$\mathrm{A} + \mathrm{NaOH} \rightleftharpoons \mathrm{B} + \mathrm{Na_2CrO_4}$$

$$\mathrm{PbSO_4} + 4\mathrm{CH_3COONH_4} \rightarrow (\mathrm{NH_4})_2\mathrm{SO_4} + \mathrm{X}$$

In the above reactions, A, B and X are respectively.

(A) $\mathrm{Na_2[Pb(OH)_2]}, \mathrm{PbCrO_4}$ and $\mathrm{[Pb(NH_3)_4]SO_4}$

(B) $\mathrm{Na_2[Pb(OH)_2]}, \mathrm{PbCrO_4}$ and $\mathrm{(NH_4)_2[Pb(CH_3COO)_4]}$

(C) $\mathrm{PbCrO_4}, \mathrm{Na_2[Pb(OH)_4]}$ and $\mathrm{(NH_4)_2[Pb(CH_3COO)_4]}$

(D) $\mathrm{PbCrO_4}, \mathrm{Na_2[Pb(OH)_4]}$ and $\mathrm{[Pb(NH_3)_4]SO_4}$

Q2. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : Among $[\mathrm{Cu(NH_3)_4}]^{2+}$, $[\mathrm{Ni(en)_3}]^{2+}$, $[\mathrm{Ni(NH_3)_6}]^{2+}$ and $[\mathrm{Mn(H_2O)_6}]^{2+}$, $[\mathrm{Mn(H_2O)_6}]^{2+}$ has the maximum number of unpaired electrons.

Statement II : The number of pairs among $\left\{[\mathrm{NiCl_4}]^{2-}, [\mathrm{Ni(CO)_4}]\right\}$, $\left\{[\mathrm{NiCl_4}]^{2-}, [\mathrm{Ni(CN)_4}]^{2-}\right\}$ and $\left\{[\mathrm{Ni(CO)_4}], [\mathrm{Ni(CN)_4}]^{2-}\right\}$ that contain only diamagnetic species is two.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are false

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

Q3. MCQ (Single Correct Answer)

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An organic compound (P) on treatment with aqueous ammonia under hot condition forms compound (Q) which on heating with $\mathrm{Br_2}$ and KOH forms compound (R) having molecular formula $\mathrm{C_6H_7N}$. Names of P, Q and R respectively are.

(A) Toluic acid, methylbenzamide, 2-methylaniline

(B) Benzoic acid, benzamide, aniline

(C) Benzoic acid, 4-methylbenzamide, 4-methylaniline

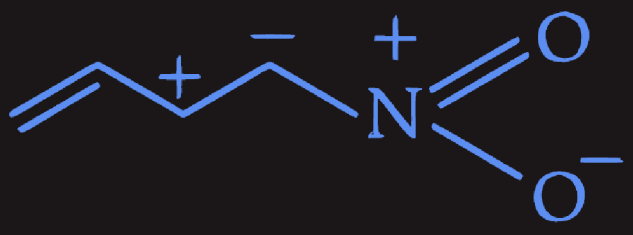
(D) Phenylethanoic acid, phenylethanamide, benzamine

Q4. MCQ (Single Correct Answer)

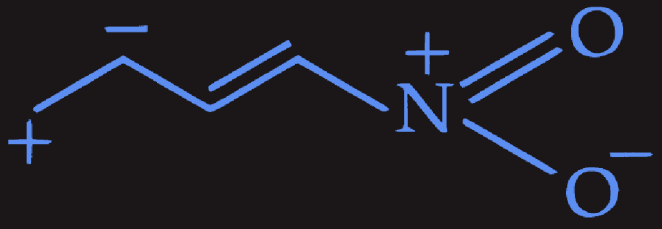
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From the following, the least stable structure is :

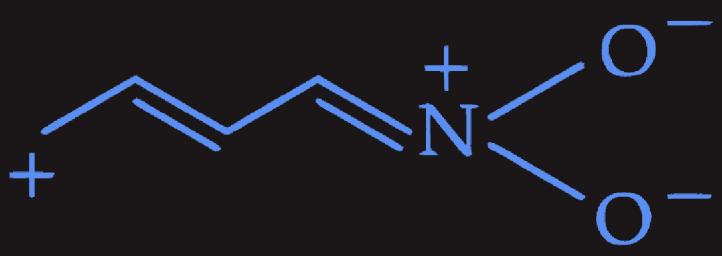
(A)



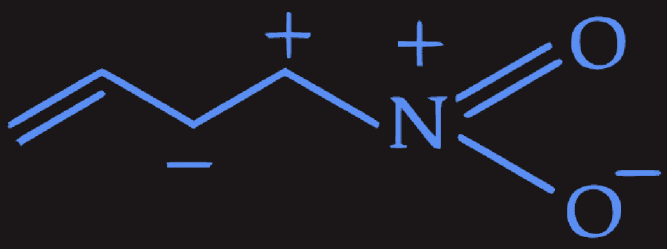
(B)



(C)



(D)



Q5. MCQ (Single Correct Answer)

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Identify the correct statements.

A. Arginine and Tryptophan are essential amino acids.

B. Histidine does not contain heterocyclic ring in its structure.

C. Proline is a six membered cyclic ring amino acid.

D. Glycine does not have chiral centre.

E. Cysteine has characteristic feature of side chain as $\mathrm{MeS-CH_2-CH_2-}$.

Choose the correct answer from the options given below :

(A) C and D Only (B) A and D Only (C) C and E Only (D) B and E Only

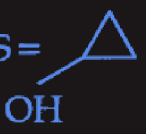
Q6. MCQ (Single Correct Answer)

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A hydrocarbon ' P ' (C_4H_8) on reaction with HCl gives an optically active compound ' Q ' (C_4H_9Cl) which on reaction with one mole of ammonia gives compound ' R ' $(C_4H_{11}N)$. ' R ' on diazotization followed by hydrolysis gives ' S '. Identify P, Q, R and S .

(A) $P=CH_3-CH=CH-CH_3,\ Q=CH_3-CH_2-CH_2-CH_2-Cl,\ R=CH_3-CH_2-CH_2-\underset{NH_2}{CH_2},\ S=CH_3-CH_2-\underset{OH}{\overset{CH_2}{CH}}$

(B) $P=CH_3-CH=CH-CH_3,\ Q=CH_3-CH_2-\underset{Cl}{CH}-CH_3,\ R=CH_3-CH_2-\underset{NH_2}{CH}-CH_3,\ S=CH_3-CH_2-\underset{OH}{CH}-CH_3$

(C) $P=CH_3$  , $Q=Cl-CH_2$  , $R=H_2N-CH_2$  , $S=$ 

(D) $P=CH_3-CH_2-CH=CH_2,\ Q=CH_3-CH_2-CH_2-\underset{Cl}{CH_2},\ R=CH_3-CH_2-CH_2-\underset{CH_3}{NH},\ S=CH_3-CH_2-\overset{OH}{\underset{CH_3}{CH}}$

Q7. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : The number of pairs among $[SiO_2, CO_2]$, $[SnO, SnO_2]$, $[PbO, PbO_2]$ and $[GeO, GeO_2]$, which contain oxides that are both amphoteric is 2 .

Statement II : BF_3 is an electron deficient molecule, can act as a Lewis acid, forms adduct with NH_3 and has a trigonal planar geometry.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

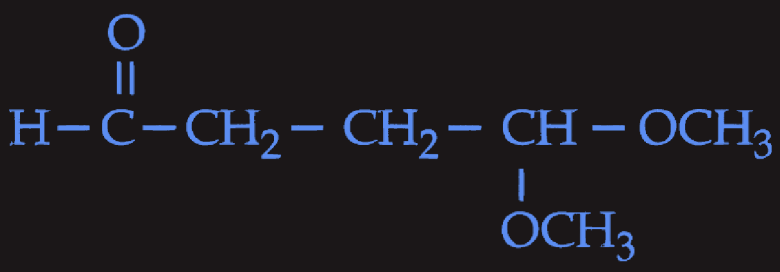
(D) Statement I is true but Statement II is false

Q8. MCQ (Single Correct Answer)

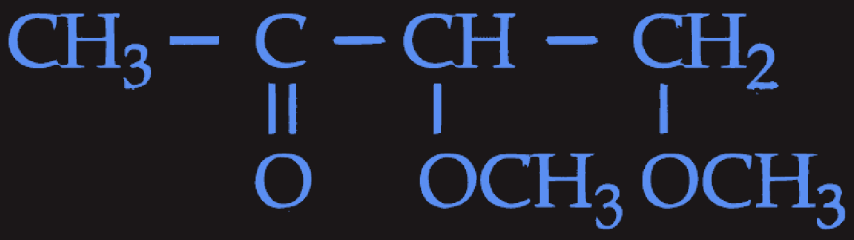
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An organic compound " P " of molecular formula $C_6H_{12}O_3$ gives positive Iodoform test but negative Tollen's test. When " P " is treated with dilute acid, it produces " Q ". " Q " gives positive Tollen's test and also iodoform test. The structure of " P " is :

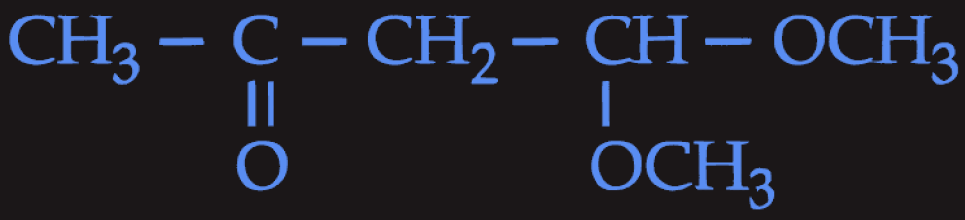
(A)



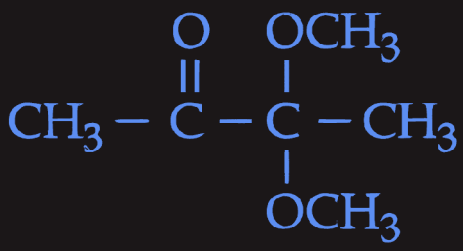
(B)



(C)



(D)

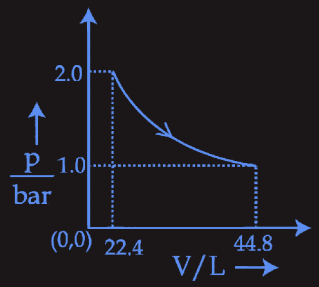


Q9. MCQ (Single Correct Answer)

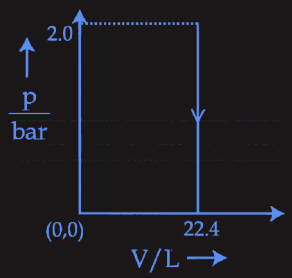
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Which of the following graphs between pressure ' p ' versus volume ' V ' represents the maximum work done?

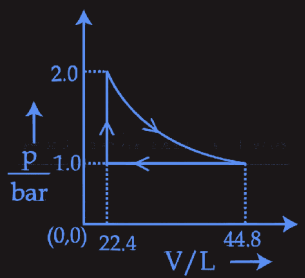
(A)



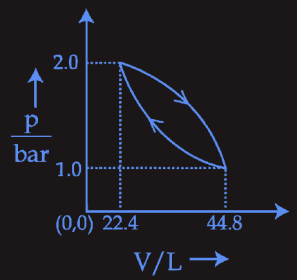
(B)



(C)



(D)



Q10. MCQ (Single Correct Answer)

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For the reaction, $N_2O_4 \rightleftharpoons 2NO_2$, graph is plotted as shown below. Identify correct statements.

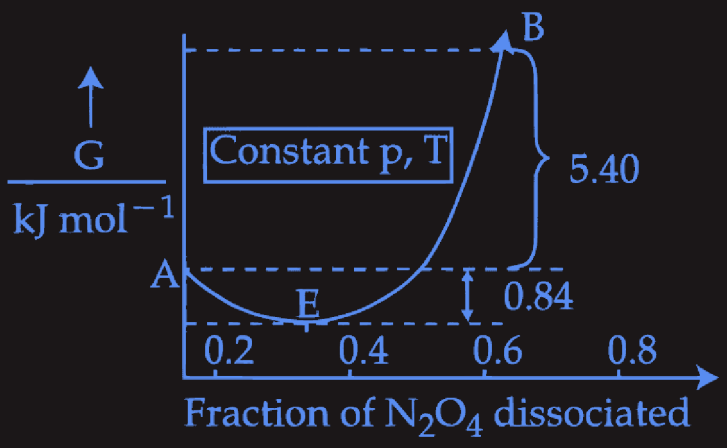
A. Standard free energy change for the reaction is $-5.40\ \text{kJ mol}^{-1}$.

B. As $\Delta G^{\ominus}$ in graph is positive, N_2O_4 will not dissociate into NO_2 at all.

C. Reverse reaction will go to completion.

D. When 1 mole of N_2O_4 changes into equilibrium mixture, value of $\Delta G^{\ominus} = -0.84\ \text{kJ mol}^{-1}$

E. When 2 mole of NO_2 changes into equilibrium mixture, $\Delta G^{\ominus}$ for equilibrium mixture is $-6.24\ \text{kJ mol}^{-1}$.



Choose the correct answer from the options given below :

(A) C and E only (B) D and E only (C) A and D only (D) B and C only

Q11. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I :

The number of species among $SF_4, NH_4^+, [NiCl_4]^{2-}, XeF_4, [PtCl_4]^{2-}, SeF_4$ and $[Ni(CN)_4]^{2-}$, that have tetrahedral geometry is 3 .

Statement II :

In the set [$NO_2, BeH_2, BF_3, AlCl_3$], all the molecules have incomplete octet around central atom.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are true

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are false

Q12. MCQ (Single Correct Answer)

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Identify **correct** statements from the following :

A. Propanal and propanone are functional isomers.

B. Ethoxyethane and methoxypropane are metamers.

C. But-2-ene shows optical isomerism.

D. But-1-ene and but-2-ene are functional isomers.

E. Pentane and 2, 2-dimethyl propane are chain isomers.

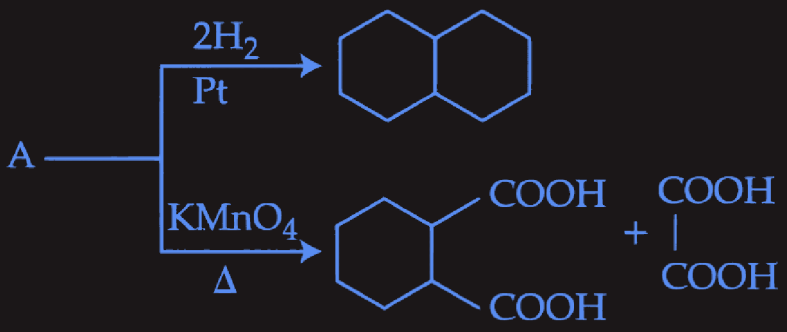
Choose the **correct** answer from the options given below :

(A) A, B and E only (B) A, B and C only (C) C, D and E only (D) B, C and D only

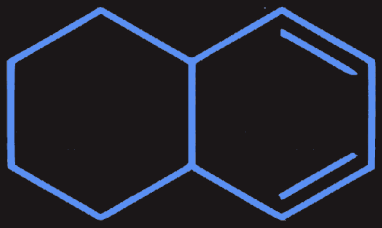
Q13. MCQ (Single Correct Answer)

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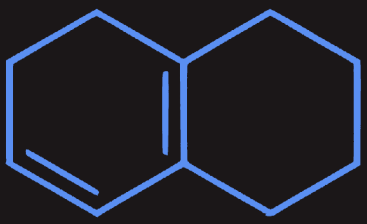
Identify A in the following reaction.



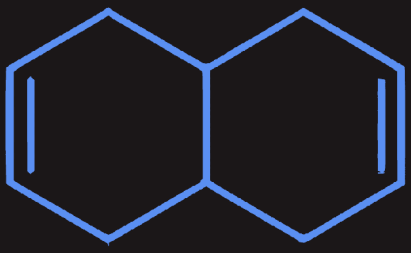
(A)



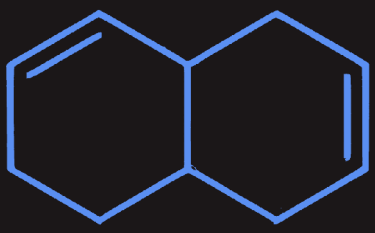
(B)



(C)



(D)



Q14. MCQ (Single Correct Answer)

14.0 g of calcium metal is allowed to react with excess HCl at 1.0 atm pressure and 273 K . Which of the following statements is incorrect?

[Given : Molar mass in gmol^{-1} of $\mathrm{Ca}-40, \mathrm{Cl}-35.5, \mathrm{H}-1$]

(A) The limiting reagent is calcium metal.

(B) 33.3 g of CaCl_2 is produced.

(C) 7.84 L of H_2 gas is evolved.

(D) 0.35 mol of H_2 gas is evolved.

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Q15. MCQ (Single Correct Answer)

Which of the following represents the correct trend for the mentioned property?

A. F > P > S > B - First Ionization Energy

B. Cl > F > S > P - Electron Affinity

C. K > Al > Mg > B - Metallic character

D. $K_2O > Na_2O > MgO > Al_2O_3$ - Basic character

Choose the **correct** answer from the options given below :

(A) B and C only

(B) A, B and D only

(C) A and B only

(D) A, B, C and D

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Q16. MCQ (Single Correct Answer)

80 mL of a hydrocarbon on mixing with 264 mL of oxygen in a closed U-tube undergoes complete combustion. The residual gases after cooling to 273 K occupy 224 mL . When the system is treated with KOH solution, the volume decreases to 64 mL . The formula of the hydrocarbon is :

(A) $\mathrm{C}_2\mathrm{H}_2$

(B) $\mathrm{C}_2\mathrm{H}_4$

(C) $\mathrm{C}_4\mathrm{H}_{10}$

(D) $\mathrm{C}_2\mathrm{H}_6$

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Q17. MCQ (Single Correct Answer)

$\mathrm{MnO_4}^{2-}$, in acidic medium, disproportionates to :

(A) MnO_4^- and MnO

(B) MnO_4^- and MnO_2

(C) $\mathrm{Mn}_2\mathrm{O}_7$ and MnO_2

(D) $\mathrm{Mn}_2\mathrm{O}_7$ and MnO

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Q18. MCQ (Single Correct Answer)

In Carius method, 0.75 g of an organic compound gave 1.2 g of barium sulphate, find percentage of sulphur (molar mass $32\ \mathrm{g\ mol}^{-1}$). Molar mass of barium sulphate is $233\ \mathrm{g\ mol}^{-1}$.

(A) 4.55%

(B) 16.48%

(C) 10.30%

(D) 21.97%

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Q19. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : When an electric discharge is passed through gaseous hydrogen, the hydrogen molecules dissociate and the energetically excited hydrogen atoms produce electromagnetic radiation of discrete frequencies.

Statement II : The frequency of second line of Balmer series obtained from He^+ is equal to that of first line of Lyman series obtained from hydrogen atom.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are false

Q20. MCQ (Single Correct Answer)

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Elements P and Q form two types of non-volatile, non-ionizable compounds PQ and PQ_2. When 1 g of PQ is dissolved in 50 g of solvent ' A', ΔT_b was 1.176 K while when 1 g of PQ_2 is dissolved in 50 g of solvent ' A', ΔT_b was 0.689 K . (K_b of ' $A' = 5\ K\ kg\ mol^{-1}$). The molar masses of elements P and Q (in $gmol^{-1}$) respectively, are :

(A) 25, 60 (B) 60, 25 (C) 65, 145 (D) 70, 110

Q21. Numerical

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Pre-exponential factors of two different reactions of same order are identical. Let activation energy of first reaction exceeds the activation energy of second reaction by $20\ kJ\ mol^{-1}$. If k_1 and k_2 are the rate constants of first and second reaction respectively at 300 K , then $\ln \frac{k_2}{k_1}$ will be ____ . (nearest integer) $\left[R = 8.3\ J\ K^{-1}\ mol^{-1}\right]$

Q22. Numerical

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Use the following data :

Substance	$\frac{\Delta_f H^\ominus (500\ K)}{kJ\ mol^{-1}}$	$\frac{S^\ominus (500\ K)}{JK^{-1}\ mol^{-1}}$
$AB(g)$	32	222
$A_2(g)$	6	146
$B_2(g)$	x	280

One mole each of $A_2(g)$ and $B_2(g)$ are taken in a 1 L closed flask and allowed to establish the equilibrium at 500 K .

$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$

The value of x $(inkJmol^{-1})$ is ____ . (Nearest integer)

(Given : $\log K = 2.2 \quad R = 8.3\ J\ K^{-1}\ mol^{-1}$)

Q23. Numerical

Consider the following reactions :

$$NaCl + K_2Cr_2O_7 + H_2SO_4 \rightarrow A + KHSO_4 + NaHSO_4 + H_2O$$
$$A + NaOH \rightarrow B + NaCl + H_2O$$
$$B + H_2SO_4 + H_2O_2 \rightarrow C + Na_2SO_4 + H_2O$$

In the product ' C ', ' X ' is the number of O_2^{2-} units, ' Y ' is the total number of oxygen atoms present and ' Z ' is the oxidation state of Cr. The value of $X + Y + Z$ is ____ .

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Q24. Numerical

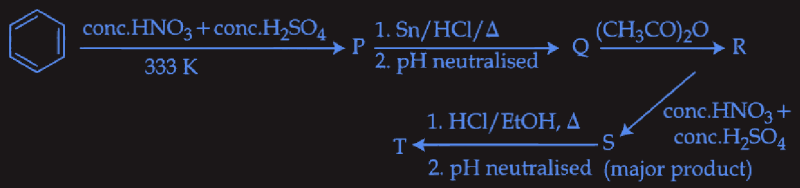
The pH and conductance of a weak acid (HX) was found to be 5 and 4×10^{-5} S, respectively. The conductance was measured under standard condition using a cell where the electrode plates having a surface area of $1\ cm^2$ were at a distance of 15 cm apart. The value of the limiting molar conductivity is ____ $Sm^2\ mol^{-1}$. (nearest integer)

(Given : degree of dissociation of the weak acid $(\alpha) \ll 1$)

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Q25. Numerical

Consider the following reaction sequence



The percentage of nitrogen in product ' T ' formed is ____ %. (Nearest integer)

(Given molar mass in $gmol^{-1}$ H : 1, C : 12, N : 14, O : 16)

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Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

Let the mean and variance of 7 observations $2, 4, 10, x, 12, 14, y, x > y$, be 8 and 16 respectively. Two numbers are chosen from $\{1, 2, 3, x-4, y, 5\}$ one after another without replacement, then the probability, that the smaller number among the two chosen numbers is less than 4 , is :

(A) $\frac{4}{5}$ (B) $\frac{3}{5}$ (C) $\frac{2}{5}$ (D) $\frac{1}{3}$

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Q27. MCQ (Single Correct Answer)

If $x^2 + x + 1 = 0$, then the value of $\left(x + \frac{1}{x}\right)^4 + \left(x^2 + \frac{1}{x^2}\right)^4 + \left(x^3 + \frac{1}{x^3}\right)^4 + \ldots + \left(x^{25} + \frac{1}{x^{25}}\right)^4$ is:

(A) 162 (B) 145 (C) 128 (D) 175

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Q28. MCQ (Single Correct Answer)

Let a point A lie between the parallel lines L_1 and L_2 such that its distances from L_1 and L_2 are 6 and 3 units, respectively. Then the area (in sq. units) of the equilateral triangle ABC , where the points B and C lie on the lines L_1 and L_2, respectively, is :

(A) $21\sqrt{3}$ (B) $12\sqrt{2}$ (C) $15\sqrt{6}$ (D) 27

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Q29. MCQ (Single Correct Answer)

The value of $\operatorname{cosec} 10^\circ - \sqrt{3}\sec 10^\circ$ is equal to :

(A) 2 (B) 6 (C) 8 (D) 4

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Q30. MCQ (Single Correct Answer)

Let $\vec{a} = -\hat{i} + 2\hat{j} + 2\hat{k}$, $\vec{b} = 8\hat{i} + 7\hat{j} - 3\hat{k}$ and $\vec{c}$ be a vector such that $\vec{a} \times \vec{c} = \vec{b}$. If $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = 4$, then $|\vec{a} + \vec{c}|^2$ is equal to :

(A) 30 (B) 33 (C) 27 (D) 35

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Q31. MCQ (Single Correct Answer)

The value of $\int_{-\pi/6}^{\pi/6} \left(\frac{\pi + 4x^{11}}{1 - \sin(|x| + \pi/6)}\right) dx$ is equal to:

(A) 8π (B) 4π (C) 2π (D) 6π

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Q32. MCQ (Single Correct Answer)

Let $y = y(x)$ be the solution curve of the differential equation $(1 + x^2)dy + (y - \tan^{-1} x)dx = 0, y(0) = 1$. Then the value of $y(1)$ is :

(A) $\frac{4}{e^{\pi/4}} - \frac{\pi}{2} - 1$ (B) $\frac{2}{e^{\pi/4}} + \frac{\pi}{4} - 1$

(C) $\frac{4}{e^{\pi/4}} + \frac{\pi}{2} - 1$ (D) $\frac{2}{e^{\pi/4}} - \frac{\pi}{4} - 1$

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Q33. MCQ (Single Correct Answer)

The number of strictly increasing functions f from the set $\{1, 2, 3, 4, 5, 6\}$ to the set $\{1, 2, 3, \ldots, 9\}$ such that $f(i) \neq i$ for $1 \leq i \leq 6$, is equal to :

(A) 21 (B) 28 (C) 27 (D) 22

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Q34. MCQ (Single Correct Answer)

Let (α, β, γ) be the co-ordinates of the foot of the perpendicular drawn from the point $(5, 4, 2)$ on the line $\vec{r} = (-\hat{i} + 3\hat{j} + \hat{k}) + \lambda(2\hat{i} + 3\hat{j} - \hat{k})$.

Then the length of the projection of the vector $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ on the vector $6\hat{i} + 2\hat{j} + 3\hat{k}$ is :

(A) $\frac{18}{7}$ (B) $\frac{15}{7}$ (C) 4 (D) 3

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Q35. MCQ (Single Correct Answer)

Let $a_1, a_2, a_3, \ldots$ be a G.P. of increasing positive terms such that $a_2 \cdot a_3 \cdot a_4 = 64$ and $a_1 + a_3 + a_5 = \frac{813}{7}$. Then $a_3 + a_5 + a_7$ is equal to :

(A) 3256 (B) 3252 (C) 3248 (D) 3244

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Q36. MCQ (Single Correct Answer)

Let PQ and MN be two straight lines touching the circle $x^2 + y^2 - 4x - 6y - 3 = 0$ at the points A and B respectively. Let O be the centre of the circle and $\angle AOB = \pi/3$. Then the locus of the point of intersection of the lines PQ and MN is :

(A) $x^2 + y^2 - 18x - 12y - 25 = 0$ (B) $x^2 + y^2 - 12x - 18y - 25 = 0$

(C) $3\left(x^2 + y^2\right) - 12x - 18y - 25 = 0$ (D) $3\left(x^2 + y^2\right) - 18x - 12y + 25 = 0$

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Q37. MCQ (Single Correct Answer)

The area of the region, inside the ellipse $x^2 + 4y^2 = 4$ and outside the region bounded by the curves $y = |x| - 1$ and $y = 1 - |x|$, is :

(A) $2\pi - 1$ (B) $3(\pi - 1)$ (C) $2(\pi - 1)$ (D) $2\pi - \frac{1}{2}$

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Q38. MCQ (Single Correct Answer)

The sum of all the roots of the equation $(x-1)^2 - 5|x-1| + 6 = 0$, is :

(A) 3 (B) 1 (C) 4 (D) 5

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Q39. MCQ (Single Correct Answer)

If the coefficient of x in the expansion of $\left(ax^2 + bx + c\right)(1 - 2x)^{26}$ is -56 and the coefficients of x^2 and x^3 are both zero, then $\mathrm{a} + \mathrm{b} + \mathrm{c}$ is equal to :

(A) 1483 (B) 1300 (C) 1500 (D) 1403

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Q40. MCQ (Single Correct Answer)

The number of relations, defined on the set $\{\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{d}\}$, which are both reflexive and symmetric, is equal to:

(A) 16 (B) 64 (C) 256 (D) 1024

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Q41. MCQ (Single Correct Answer)

Let the foci of a hyperbola coincide with the foci of the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$. If the eccentricity of the hyperbola is 5 , then the length of its latus rectum is :

(A) $\frac{96}{\sqrt{5}}$ (B) $24\sqrt{5}$ (C) 12 (D) 16

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Q42. MCQ (Single Correct Answer)

If the domain of the function $f(x)=\cos^{-1}\left(\frac{2x-5}{11-3x}\right)+\sin^{-1}\left(2x^2-3x+1\right)$ is the interval $[\alpha,\beta]$, then $\alpha+2\beta$ is equal to :

(A) 5 (B) 2 (C) 3 (D) 1

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Q43. MCQ (Single Correct Answer)

Let O be the vertex of the parabola $x^2=4y$ and Q be any point on it. Let the locus of the point P , which divides the line segment OQ internally in the ratio $2:3$ be the conic C . Then the equation of the chord of C, which is bisected at the point $(1,2)$, is :

(A) $5x-4y+3=0$ (B) $x-2y+3=0$ (C) $4x-5y+6=0$ (D) $5x-y-3=0$

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Q44. MCQ (Single Correct Answer)

Let $\overrightarrow{c}$ and $\overrightarrow{\mathrm{d}}$ be vectors such that $|\overrightarrow{c}+\overrightarrow{\mathrm{d}}|=\sqrt{29}$ and $\overrightarrow{c}\times(2\hat{i}+3\hat{j}+4\hat{k})=(2\hat{i}+3\hat{j}+4\hat{k})\times\overrightarrow{\mathrm{d}}$. If $\lambda_1,\lambda_2\,(\lambda_1>\lambda_2)$ are the possible values of $(\vec{c}+\vec{d})\cdot(-7\hat{i}+2\hat{j}+3\hat{k})$, then the equation $\mathrm{K}^2x^2+\left(\mathrm{K}^2-5\,\mathrm{K}+\lambda_1\right)xy+\left(3\,\mathrm{K}+\frac{\lambda_2}{2}\right)y^2-8x+12y+\lambda_2=0$ represents a circle, for K equal to :

(A) 4 (B) -1 (C) 2 (D) 1

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Q45. MCQ (Single Correct Answer)

Let $f:\mathbf{R}\to(0,\infty)$ be a twice differentiable function such that $f(3)=18, f'(3)=0$ and $f''(3)=4$.

Then $\lim\limits_{x\to1}\left(\log_e\left(\frac{f(2+x)}{f(3)}\right)^{\frac{18}{(x-1)^2}}\right)$ is equal to :

(A) 9 (B) 18 (C) 1 (D) 2

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Q46. Numerical

Let $f:\mathbf{R}\to\mathbf{R}$ be a twice differentiable function such that the quadratic equation $f(x)\mathrm{m}^2-2f'(x)\mathrm{m}+f''(x)=0$ in m , has two equal roots for every $x\in\mathbf{R}$. If $f(0)=1, f'(0)=2$, and (α,β) is the largest interval in which the function $f\left(\log_e x-x\right)$ is increasing, then $\alpha+\beta$ is equal to

____ .

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Q47. Numerical

Let $S=\{(m,n):m,n\in\{1,2,3,\ldots\ldots,50\}\}$. If the number of elements (m,n) in S such that 6^m+9^n is a multiple of 5 is p and the number of elements (m,n) in S such that $m+n$ is a square of a prime number is q , then $\mathrm{p}+\mathrm{q}$ is equal to ____ .

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Q48. Numerical

Let $a_1 = 1$ and for $n \geqslant 1, a_{n+1} = \frac{1}{2}a_n + \frac{n^2-2n-1}{n^2(n+1)^2}$. Then $\left|\sum_{n=1}^{\infty}\left(a_n - \frac{2}{n^2}\right)\right|$ is equal to ____ .

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Q49. Numerical

$6\int_0^{\pi}|(\sin 3x + \sin 2x + \sin x)|dx$ is equal to ____ .

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Q50. Numerical

For some $\alpha, \beta \in \mathbf{R}$, let $A = \begin{bmatrix}\alpha & 2\\ 1 & 2\end{bmatrix}$ and $B = \begin{bmatrix}1 & 1\\ 1 & \beta\end{bmatrix}$ be such that $A^2 - 4A + 2I = B^2 - 3B + I = O$. Then $\left(\det\left(\text{adj}\left(A^3 - B^3\right)\right)\right)^2$ is equal to ____ .

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Physics 25 questions

Q51. MCQ (Single Correct Answer)

A conducting circular loop of area $1.0\,\text{m}^2$ is placed perpendicular to a magnetic field which varies as $B = \sin(100t)$ Tesla. If the resistance of the loop is 100Ω, then the average thermal energy dissipated in the loop in one period is ____ J.

(A) π^2 (B) $\frac{\pi}{2}$ (C) π (D) 2π

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Q52. MCQ (Single Correct Answer)

A 4 kg mass moves under the influence of a force $\vec{F} = \left(4t^3\hat{i} - 3t\hat{j}\right)\text{N}$ where t is the time in second. If mass starts from origin at $t = 0$, the velocity and position after $t = 2\,\text{s}$ will be:

(A) $\vec{v} = 3\hat{i} + \frac{3}{2}\hat{j} \quad \vec{r} = \frac{6}{5}\hat{i} + \hat{j}$

(B) $\vec{v} = 4\hat{i} - \frac{3}{2}\hat{j} \quad \vec{r} = \frac{6}{5}\hat{i} - \hat{j}$

(C) $\vec{v} = 4\hat{i} - \frac{3}{2}\hat{j} \quad \vec{r} = \frac{8}{5}\hat{i} - \hat{j}$

(D) $\vec{v} = 4\hat{i} + \frac{5}{2}\hat{j} \quad \vec{r} = \frac{8}{5}\hat{i} + 2\hat{j}$

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Q53. MCQ (Single Correct Answer)

In an experiment the values of two spring constants were measured as $k_1 = (10 \pm 0.2)\text{N/m}$ and $k_2 = (20 \pm 0.3)\text{N/m}$. If these springs are connected in parallel, then the percentage error in equivalent spring constant is :

(A) 1.33% (B) 2.67% (C) 1.67% (D) 2.33%

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Q54. MCQ (Single Correct Answer)

Consider a modified Bernoulli equation.

$\left(\mathrm{P}+\frac{A}{Bt^2}\right)+\rho g(h+Bt)+\frac{1}{2}\rho V^2=$ constant

If t has the dimension of time then the dimensions of A and B are ____ , ____ respectively.

(A) $\left[\mathrm{ML^0\,T^{-1}}\right]$ and $\left[\mathrm{M^0LT}\right]$

(B) $\left[\mathrm{ML^0\,T^{-2}}\right]$ and $\left[\mathrm{M^0LT^{-1}}\right]$

(C) $\left[\mathrm{ML^0\,T^{-2}}\right]$ and $\left[\mathrm{M^0LT^{-2}}\right]$

(D) $\left[\mathrm{ML^0\,T^{-1}}\right]$ and $\left[\mathrm{M^0LT^{-1}}\right]$

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Q55. MCQ (Single Correct Answer)

An aluminium and steel rods having same lengths and cross-sections are joined to make total length of 120 cm at $30^\circ\mathrm{C}$. The coefficient of linear expansion of aluminium and steel are $24\times10^{-6}/^\circ\mathrm{C}$ and $1.2\times10^{-5}/^\circ\mathrm{C}$, respectively. The length of this composite rod when its temperature is raised to $100^\circ\mathrm{C}$, is ____ cm.

(A) 120.20 (B) 120.06 (C) 120.15 (D) 120.03

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Q56. MCQ (Single Correct Answer)

Two strings (A, B) having linear densities $\mu_A=2\times10^{-4}\,\mathrm{kg/m}$ and, $\mu_B=4\times10^{-4}\,\mathrm{kg/m}$ and lengths $L_A=2.5\,\mathrm{m}$ and $L_B=1.5\,\mathrm{m}$ respectively are joined. Free ends of A and B are tied to two rigid supports C and D, respectively creating a tension of 500 N in the wire. Two identical pulses, sent from C and D ends, take time t_1 and t_2, respectively, to reach the joint. The ratio t_1/t_2 is:

(A) 1.90 (B) 1.18 (C) 1.08 (D) 1.67

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Q57. MCQ (Single Correct Answer)

The electric field in a plane electromagnetic wave is given by :

$E_y=69\sin\left[0.6\times10^3x-1.8\times10^{11}t\right]\mathrm{V/m}.$

The expression for magnetic field associated with this electromagnetic wave is ____ T.

(A) $B_z=2.3\times10^{-7}\sin\left[0.6\times10^3x-1.8\times10^{11}t\right]$

(B) $B_z=2.3\times10^{-7}\sin\left[0.6\times10^3x+1.8\times10^{11}t\right]$

(C) $B_y=2.3\times10^{-7}\sin\left[0.6\times10^3x-1.8\times10^{11}t\right]$

(D) $B_y=69\sin\left[0.6\times10^3x+1.8\times10^{11}t\right]$

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Q58. MCQ (Single Correct Answer)

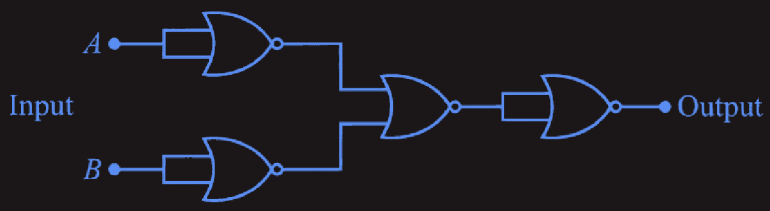
A parallel plate capacitor has capacitance C, when there is vacuum within the parallel plates. A sheet having thickness $\left(\frac{1}{3}\right)^{\text{rd}}$ of the separation between the plates and relative permittivity K is introduced between the plates. The new capacitance of the system is :

(A) $\frac{3CK^2}{(2K+1)^2}$ (B) $\frac{4KC}{3K-1}$ (C) $\frac{CK}{2+K}$ (D) $\frac{3KC}{2K+1}$

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Q59. MCQ (Single Correct Answer)

The given circuit works as :



(A) NOR gate (B) OR gate (C) AND gate (D) NAND gate

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Q60. MCQ (Single Correct Answer)

A point charge of 10^{-8}C is placed at origin. The work done in moving a point charge $2\mu\text{C}$ from point $A(4,4,2)\text{m}$ to $B(2,2,1)\text{m}$ is ____ J. $\left(\frac{1}{4\pi\epsilon_o} = 9\times10^9 \text{ in SI units}\right)$

(A) 30×10^{-6} (B) 0 (C) 15×10^{-6} (D) 45×10^{-6}

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Q61. MCQ (Single Correct Answer)

A light wave described by $E = 60\left[\sin\left(3\times10^{15}\right)t + \sin\left(12\times10^{15}\right)t\right]$ (in SI units) falls on a metal surface of work function 2.8 eV . The maximum kinetic energy of ejected photoelectron is (approximately)

____ eV. $\left(h = 6.6\times10^{-34}\text{ J.s. and } e = 1.6\times10^{-19}\text{C}\right)$

(A) 3.8 (B) 7.8 (C) 6.0 (D) 5.1

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Q62. MCQ (Single Correct Answer)

Initially a satellite of 100 kg is in a circular orbit of radius 1.5R_E. This satellite can be moved to a circular orbit of radius $3R_E$ by supplying $\alpha\times10^6$ J of energy. The value of α is ____ .

(Take Radius of Earth $R_E = 6\times10^6$ m and $\text{g} = 10\,\text{m/s}^2$)

(A) 500 (B) 1000 (C) 100 (D) 150

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Q63. MCQ (Single Correct Answer)

A gas based geyser heats water flowing at the rate of 5.0 litres per minute from $27^{\circ}\mathrm{C}$ to $87^{\circ}\mathrm{C}$. The rate of consumption of the gas is ____ $\mathrm{g/s}$.

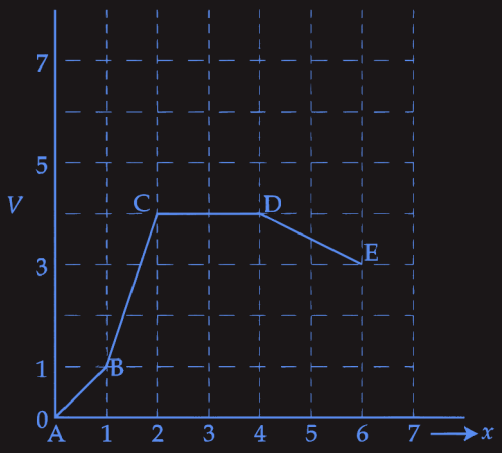
(Take heat of combustion of gas $= 5.0 \times 10^4\ \mathrm{J/g}$, specific heat capacity of water $= 4200\ \mathrm{J/kg \cdot {}^{\circ}C}$)

(A) 4.2 (B) 2.1 (C) 0.21 (D) 0.42

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Q64. MCQ (Single Correct Answer)

Potential energy (V) versus distance (x) is given by the graph. Rank various regions as per the magnitudes of the force (F) acting on a particle from high to low.



(A) $F_{BC} > F_{AB} > F_{DE} > F_{CD}$ (B) $F_{BC} > F_{CD} > F_{DE} > F_{AB}$

(C) $F_{CD} > F_{AB} > F_{BC} > F_{DE}$ (D) $F_{CD} > F_{DE} > F_{AB} > F_{BC}$

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Q65. MCQ (Single Correct Answer)

If an alpha particle with energy 7.7 MeV is bombarded on a thin gold foil, the closest distance from nucleus it can reach is ____ m. (Atomic number of gold $= 79$ and $\frac{1}{4\pi\epsilon_o} = 9 \times 10^9$ in SI units)

(A) 2.95×10^{-16} (B) 3.85×10^{-14} (C) 2.95×10^{-14} (D) 3.85×10^{-16}

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Q66. MCQ (Single Correct Answer)

A current carrying solenoid is placed vertically and a particle of mass m with charge Q is released from rest. The particle moves along the axis of solenoid. If g is acceleration due to gravity then the acceleration (a) of the charged particle will satisfy :

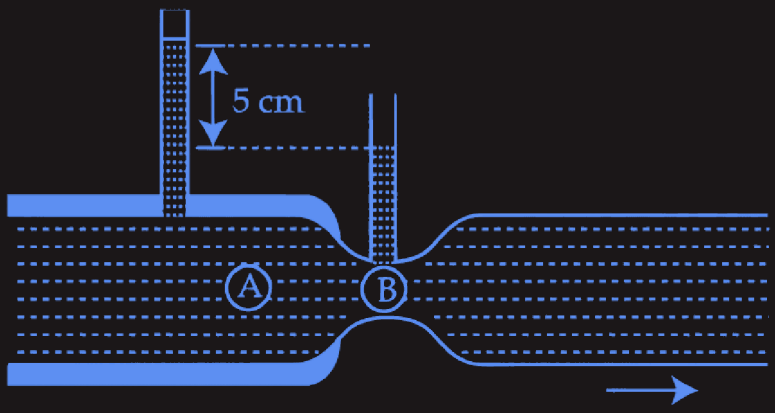
(A) $a > g$ (B) $a = 0$ (C) $0 < a < g$ (D) $a = g$

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Q67. MCQ (Single Correct Answer)

Water flows through a horizontal tube as shown in the figure. The difference in height between the water columns in vertical tubes is 5 cm and the area of cross-sections at A and B are $6\ \mathrm{cm}^2$ and $3\ \mathrm{cm}^2$ respectively. The rate of flow will be ____ cm^3/s. (take $g = 10\ \mathrm{m/s}^2$)

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(A) $100\sqrt{3}$ (B) $\frac{200}{\sqrt{3}}$ (C) $200\sqrt{6}$ (D) $200\sqrt{3}$

Q68. MCQ (Single Correct Answer)

In a double slit experiment the distance between the slits is 0.1 cm and the screen is placed at 50 cm from the slits plane. When one slit is covered with a transparent sheet having thickness t and refractive index $n(=1.5)$, the central fringe shifts by 0.2 cm . The value of t is

____ cm.

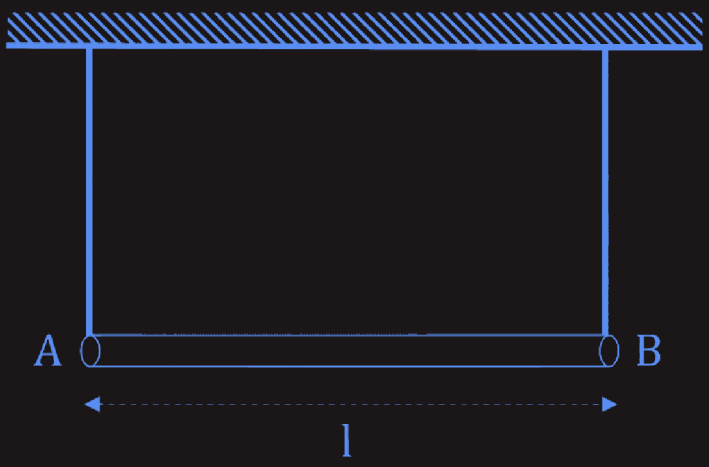
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(A) 6.0×10^{-3} (B) 8×10^{-4} (C) 5.0×10^{-3} (D) 5.6×10^{-4}

Q69. MCQ (Single Correct Answer)

A uniform rod of mass m and length l is suspended by means of two identical inextensible light strings as shown in figure. Tension in one string immediately after the other string is cut, is ____ . (g acceleration due to gravity)

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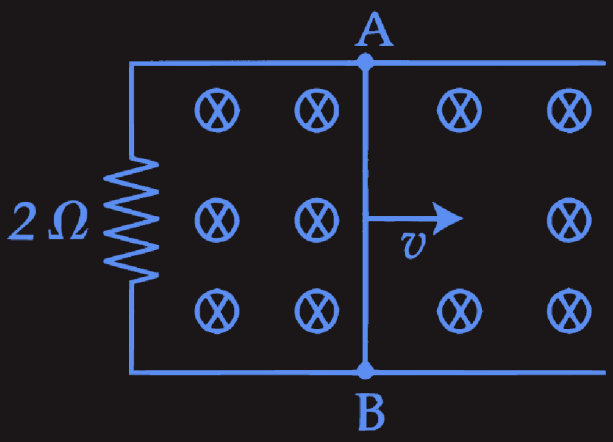


(A) $\mathrm{mg/s}$ (B) $mg/4$ (C) mg (D) $mg/2$

Q70. MCQ (Single Correct Answer)

A 1 m long metal rod AB completes the circuit as shown in figure. The area of circuit is perpendicular to the magnetic field of 0.10 T . If the resistance of the total circuit is 2Ω then the force needed to move the rod towards right with constant speed (v) of $1.5\ \mathrm{m/s}$ is ____ N.

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(A) 5.7×10^{-2} (B) 7.5×10^{-3} (C) 5.7×10^{-3} (D) 7.5×10^{-2}

Q71. Numerical

10 mole of oxygen is heated at constant volume from $30^{\circ}\mathrm{C}$ to $40^{\circ}\mathrm{C}$. The change in the internal energy of the gas is ____ cal. (The molecular specific heat of oxygen at constant pressure, $C_P = 7\mathrm{cal/mol.}\,^{\circ}\mathrm{C}$ and $\mathrm{R} = 2\mathrm{cal.}/\mathrm{mol.}\,^{\circ}\mathrm{C.}$)

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Q72. Numerical

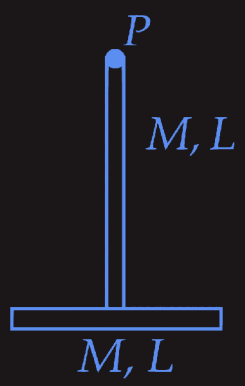
In a microscope the objective is having focal length $f_o = 2\ \mathrm{cm}$ and eye-piece is having focal length $f_e = 4\ \mathrm{cm}$. The tube length is 32 cm . The magnification produced by this microscope for normal adjustment is ____ .

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Q73. Numerical

Two identical thin rods of mass M kg and length L m are connected as shown in figure. Moment of inertia of the combined rod system about an axis passing through point P and perpendicular to the plane of the rods is $\frac{x}{12}ML^2$ kg m^2. The value of x is ____ .

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Q74. Numerical

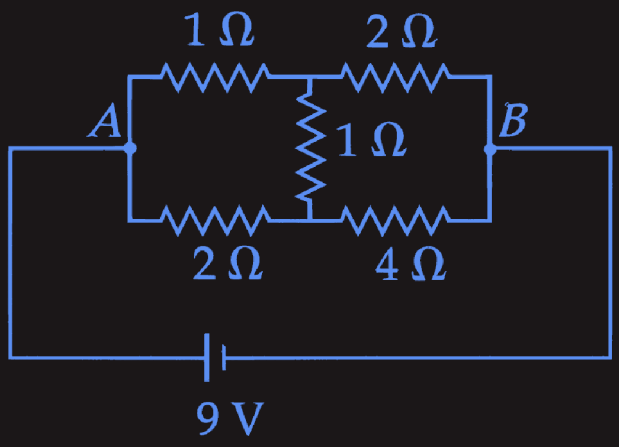
A collimated beam of light of diameter 2 mm is propagating along x-axis. The beam is required to be expanded in a collimated beam of diameter 14 mm using a system of two convex lenses. If first lens has focal length 40 mm , then the focal length of second lens is ____ mm.

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Q75. Numerical

The heat generated in 1 minute between points A and B in the given circuit, when a battery of 9 V with internal resistance of 1Ω is connected across these points is ____ J.

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Answer Key

Chemistry

1. **C**	2. **D**	3. **B**	4. **D**	5. **B**
6. **B**	7. **C**	8. **C**	9. **A**	10. **B**
11. **C**	12. **A**	13. **A**	14. **B**	15. **B**
16. **A**	17. **B**	18. **D**	19. **A**	20. **A**
21. **8**	22. **70**	23. **13**	24. **6**	25. **20**

Mathematics

26. **A**	27. **B**	28. **A**	29. **D**	30. **C**
31. **B**	32. **B**	33. **B**	34. **A**	35. **B**
36. **C**	37. **C**	38. **C**	39. **D**	40. **B**
41. **A**	42. **C**	43. **A**	44. **D**	45. **D**
46. **1**	47. **1333**	48. **2**	49. **17**	50. **225**

Physics

51. **C**	52. **C**	53. **C**	54. **D**	55. **C**
56. **B**	57. **A**	58. **D**	59. **D**	60. **A**
61. **D**	62. **B**	63. **D**	64. **A**	65. **C**
66. **D**	67. **D**	68. **B**	69. **B**	70. **B**
71. **500**	72. **100**	73. **17**	74. **280**	75. **1080**

For numerical answers given as a range, any value within the range is correct.

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